

SHARJEEL SAFDAR

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PROFESSIONAL SUMMARY

Computer Science graduate specializing in Machine Learning, Retrieval-Augmented Generation (RAG), and Data Science, with hands-on experience across a data science internship and project-based AI engineering. Skilled in Python, FastAPI, React, LLM integration, feature engineering, and end-to-end model deployment. Experienced in leading research-driven software projects using Agile/Scrum.

EXPERIENCE

Data Science Intern | 10Pearls

Jul 2026 – Sep 2026

Project: Air Quality Index (AQI) Forecasting | Karachi, PK

- Built an end-to-end AQI forecasting pipeline for **3 cities** (Karachi, Lahore, Islamabad), consolidating **4 years** of hourly Open-Meteo data (**211,824** raw records) into **105,912** validated records with **0** duplicate or missing timestamps.
- Cleaned and validated **11** air-quality and weather variables in Python (**0.137%** missing-value rate) and engineered **354 features**, including lags up to 168 hours, rolling statistics, and seasonal interactions, passing **6/6** data-leakage checks.
- Performed exploratory data analysis (EDA) and SHAP analysis to identify the strongest AQI drivers; recent AQI peaks and sustained high-pollution hours were the top predictors of 24-hour forecasts.
- Benchmarked **3 ML models against 2 baselines** and selected Random Forest (validation RMSE **22.85** vs. 27.58 for persistence, about **17% lower**), reaching $R^2 = 0.82$ at 24 hours and **0.73** overall on a held-out test set.
- Delivered 24/48/72-hour forecasts through a FastAPI and Next.js dashboard, automated with **3 GitHub Actions pipelines** (hourly and daily) and validated with **58 Python and 24 frontend tests**.

AI Engineer – Project-Based | Generation Next Technologies

Apr 2026 – Jun 2026

Project: Snakitos RAG | Karachi, PK

- Developed a Hybrid Retrieval-Augmented Generation (RAG) system for Snakitos, using a **Pinecone vector database** to retrieve relevant context and ground LLM answers in Snakitos data.
- Integrated LLM APIs with a custom Pinecone-backed retrieval backend, delivering intelligent, context-aware responses to end users.
- Collaborated with the engineering team to test and iterate on the RAG pipeline across varied query types, refining query handling and response quality.
- Evaluated retrieval and generation through structured testing, applying iterative improvements to the pipeline.

PROJECTS

Personalized Crypto — Final Year Project | Project Manager & Team Lead (Team of 3) personalizedcrypto2025.vercel.app

- Led a **3-member team** as Project Manager and Group Leader, delivering an AI-powered crypto trading-education platform with **9 integrated modules** (live market grid, paper trading, HODL and DCA simulators, converter, assessment, AI learning path, learning hub).
- Led research and design: reviewed **34 studies across 6 themes** to guide AI/ML architecture and feature-engineering decisions, and co-authored the project research paper.
- Trained and compared **3 ML models** on a **10,000-record** trader-profile dataset (SMOTE-balanced, 80/20 split): Random Forest classifier **89.00%** accuracy, Gradient Boosting R^2 **85.85%**, Polynomial Regression risk model R^2 **86.89%** (MAE 5.48).
- Engineered behavioral features (Win Rate, Loss Ratio, Knowledge Deficit, Capital Exposure Index) and **66 interaction terms**, raising risk-model R^2 from about **70% to 86.89%** and outperforming a deep-learning baseline (72.8–79.7%).
- Built the full-stack platform with React 18/Vite, FastAPI, and Supabase (PostgreSQL, JWT, Row-Level Security); integrated the CoinGecko API (top **100** coins, **30-second** refresh), a **150-question** randomized quiz bank, and GPT-4o-mini for personalized learning roadmaps and PDF reports; deployed on Vercel.
- Managed delivery with Agile/Scrum: sprint planning, backlog prioritization, and blocker resolution across the 3-person team.

AI Interview Coach

- Built an adaptive interview platform using React, FastAPI, SQLite, and the Web Speech API.
- Developed resume-based interview question generation, voice analytics, AI scoring, and model-answer comparison.
- Implemented persistent reports and performance tracking with Ollama/Claude integration and Docker deployment.

EDUCATION

FAST University — Bachelor of Computer Science

Aug 2022 – Jun 2026

Relevant Coursework: Artificial Intelligence, Software Engineering, Software Project Management, DevOps, Software Testing, Object-Oriented Programming

Bahria College Karsaz — Intermediate, Pre-Engineering (Grade A+)

May 2017 – May 2022

TECHNICAL SKILLS

Programming: Python, Java, JavaScript
AI / Machine Learning: Machine Learning, Retrieval-Augmented Generation (RAG), LLM Integration, Pinecone (Vector Database), Prompt Engineering, Scikit-learn, TensorFlow/Keras, SHAP, Random Forest, Gradient Boosting, Feature Engineering, Exploratory Data Analysis (EDA), Data Preprocessing
Web & Backend: React, Next.js, Vite, FastAPI, REST APIs, Supabase, PostgreSQL, SQLite
DevOps & Tools: Docker, GitHub Actions (CI/CD), Git, Linux, Vercel, Render, Hopsworks, Jira, pytest, VS Code
Software Engineering: Agile/Scrum, Software Development Lifecycle (SDLC), Software Testing, Sprint Planning, Backlog Management

ADDITIONAL SKILLS

Research & Management: Research and Development (R&D), Technical Research, Requirements Analysis, Risk Management, Cross-functional Collaboration, Team Leadership
Digital Marketing: Social Media Marketing, TikTok Marketing, Snapchat Marketing, Content Strategy, Campaign Planning

LANGUAGES

Urdu (Native) | English (Professional Proficiency)